THE LOCUS OF GEOMETRICALLY REDUCED FIBRES IS LOCALLY CONSTRUCTIBLE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 7258-7266 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 7268-7302, the complete author proof (33 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Geometric reducedness of fibres, constructibility of finite-presentation conditions, and base change to an algebraic closure; the reduction is retained.

Source reference. Original lemma lemma-base-change-fibres-geometrically-reduced: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7161–7169. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-geometrically-reduced-generic-fibre: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7215–7222. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-not-geometrically-reduced-in-neighbourhood: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7181–7186. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01ZM: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7302–7302. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 053Z: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7302–7302. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a morphism which is quasi-compact and locally of finite presentation. Then the set*

$$E = \{y \in Y \mid X_y \text{ is geometrically reduced}\}$$

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is locally constructible in Y.

Proof. Let $y \in Y$. We have to show that there exists an open neighbourhood V of y in Y such that $E \cap V$ is constructible in V. Thus we may assume that Y is affine. Then X is quasi-compact. Choose a finite affine open covering $X = U_1 \cup \ldots \cup U_n$. Then the fibres of $U_i \to Y$ at y form an affine open covering of the fibre of $X \to Y$ at y. Hence we may assume X is affine as well. Write $Y = \mathrm{Spec}(A)$. Write $A = \mathrm{colim}\, A_i$ as a directed limit of finite type $\mathbf{Z}$-algebras. By Limits, Lemma 01ZM we can find an i and a morphism $f_i : X_i \to \mathrm{Spec}(A_i)$ of finite presentation whose base change to Y recovers f. By Lemma lemma-base-change-fibres-geometrically-reduced it suffices to prove the lemma for f_i. Thus we reduce to the case where Y is the spectrum of a Noetherian ring.

We will use the criterion of Topology, Lemma 053Z to prove that E is constructible in case Y is a Noetherian scheme. To see this let $Z \subset Y$ be an irreducible closed subscheme with generic point ξ. We have to show that $E \cap Z$ either contains a nonempty open subset or is not dense in Z. If X_ξ is geometrically reduced, then Lemma lemma-geometrically-reduced-generic-fibre (applied to the morphism $X_Z \to Z$) implies that all fibres X_y are geometrically reduced for a nonempty open $V \subset Z$. If X_ξ is not geometrically reduced, then Lemma lemma-not-geometrically-reduced-in-neighbourhood (applied to the morphism $X_Z \to Z$) implies that all fibres X_y are not geometrically reduced for a nonempty open $V \subset Z$. Thus we win. $\square$